

Optimization 1

ISE 5406

Lecture 6: KKT Conditions & Constrained Optimization

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Where We Are in the Course

Last Time (Lecture 5): Subgradients and Unconstrained Optimization

- Separating (Thm. 5.3) and supporting hyperplane theorems (Thm. 5.5)
- Subgradients and subdifferentials
- Local optimality $\Rightarrow$ global optimality for convex problems (Thm. 5.6)
- Optimality conditions: $\mathbf{0} \in \partial f(\bar{\mathbf{x}})$ for unconstrained problems (Thm. 5.7, Cor. 5.8)

Today (Lecture 6): Constrained Optimization and KKT Conditions

- The Lagrangian function and Lagrange multipliers
- Karush-Kuhn-Tucker (KKT) necessary conditions
- KKT sufficient conditions for convex problems
- Constraint qualification (Slater's condition)
- Geometric interpretation of KKT conditions

Next Time (Lecture 7): Duality Theory

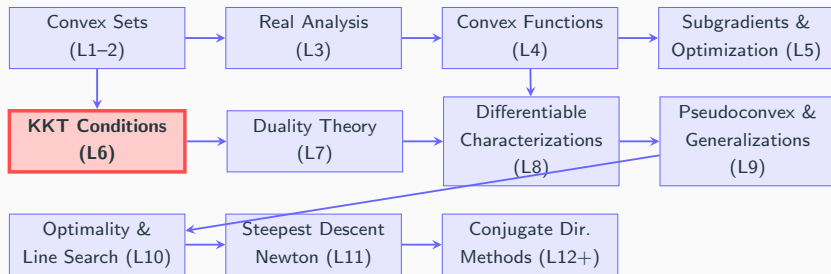
- The Lagrangian dual problem
- Weak and strong duality
- Economic interpretation of dual variables

Coming Soon (Lecture 8): Algorithms for Convex Optimization

- Gradient descent and convergence analysis
- Projected gradient methods
- Interior point methods introduction

Big Picture: KKT conditions are the bridge from theory to algorithms—they tell us *what* an optimal solution looks like, which guides algorithm design.

Big Picture: Course Roadmap



Why KKT Conditions Matter

- Provide necessary conditions for constrained optimization
- For convex problems, KKT conditions are also sufficient
- Foundation for Lagrangian duality and algorithm design

Reading for This Lecture

Course Notes (Primary References)

- Chapter 15: KKT Conditions for Inequality Constraints
- Chapter 16: KKT Conditions for Mixed Constraints
- Chapter 5: Convex Analysis (for supporting theorems on convexity)

Textbook References (Supplementary)

- Section 4.3 of Bazaraa et al. (2006)
- Sections 5.1, 5.2, and 5.5 of Boyd and Vandenberghe

Key Topics

- Lagrangian function and multipliers (Def. 15.1)
- KKT conditions: necessary and sufficient (Thm. 15.7, 15.11)
- Constraint qualifications: LICQ (Def. 15.7) and Slater's condition (Def. 15.11)

Outline

The Constrained Optimization Problem

The Lagrangian Function

KKT Conditions

- KKT Necessary Conditions

- KKT Sufficient Conditions

Constraint Qualifications

Review: Calculus Tools for KKT Proofs

Proof Space: Key Results

Applications and Examples

Warm-Up: How Do Constraints Affect Optimization?

Think About It

Find the optimal solution and optimal value for each problem. How and why does the answer change as we add constraints?

1. $\min_{x,y} x^2 + y^2$

2. $\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad x + y \geq 1$

3. $\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad x + y \geq 1, \quad x \geq 0.75$

4. $\min_{x,y} (x - 2)^2 + (y - 2)^2 \quad \text{s.t.} \quad x^2 + y^2 \leq 1$

Hint: Think geometrically. What do the level curves look like? Where is the feasible region?

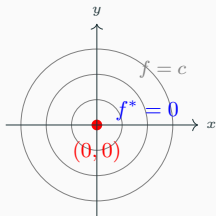
Solution: Warm-Up (a) and (b)

(a) Unconstrained

$$\min_{x,y} x^2 + y^2$$

$$\mathbf{x}^* = (0, 0), \quad f^* = 0$$

$\nabla f = (2x, 2y) = \mathbf{0}$ at the origin.

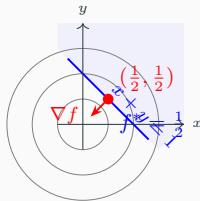


(b) One constraint

$$\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad x + y \geq 1$$

$$\mathbf{x}^* = \left(\frac{1}{2}, \frac{1}{2}\right), \quad f^* = \frac{1}{2}$$

Constraint pushes us away from $(0, 0)$.



Observation: Adding a constraint increased f^* from 0 to $\frac{1}{2}$. At the optimum, ∇f points into the *infeasible* region—we'd like to go there but can't!

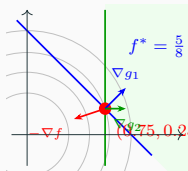
Solution: Warm-Up (c) and (d)

(c) Two constraints

$$\min x^2 + y^2 \quad \text{s.t. } x + y \geq 1, \quad x \geq 0.75$$

$$\mathbf{x}^* = (0.75, 0.25), \quad f^* = \frac{5}{8}$$

Both constraints are **active** at the optimum.



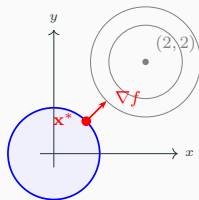
(d) Nonlinear constraint

$$\min (x-2)^2 + (y-2)^2$$

$$\text{s.t. } x^2 + y^2 \leq 1$$

$$\mathbf{x}^* = \frac{1}{\sqrt{2}}(1, 1),$$
$$f^* = (2\sqrt{2} - 1)^2$$

Closest feasible point to $(2, 2)$.



Key: $-\nabla f$ is a nonneg. combination of the active constraint gradients. Today we formalize this!

What Do These Examples Tell Us?

Key Insights from the Warm-Up

1. **Constraints can only increase (or maintain) the optimal value.**

$$(a) f^* = 0 \rightarrow (b) f^* = 1/2 \rightarrow (c) f^* = 5/8$$

2. **At a constrained optimum, ∇f points into the infeasible region.**

You cannot move feasibly to reduce f .

3. **$-\nabla f$ is a nonnegative combination of the active constraint gradients.**

Inactive constraints play no role (their weights are zero).

4. **These weights measure sensitivity.**

How much does f^* change if we tighten or relax a constraint?

Today's Agenda

We formalize these observations into the **Karush–Kuhn–Tucker (KKT) conditions** and develop the **Lagrangian** framework that makes these ideas precise.

The Constrained Optimization Problem

The Lagrangian Function

KKT Conditions

KKT Necessary Conditions

KKT Sufficient Conditions

Constraint Qualifications

Review: Calculus Tools for KKT Proofs

Proof Space: Key Results

Applications and Examples

Standard Form: Constrained Optimization

Consider the general **constrained optimization problem**:

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\ & h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p \\ & \mathbf{x} \in \mathcal{X} \end{aligned} \tag{1}$$

where:

- $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the **objective function**
- $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ are **inequality constraints**
- $h_j : \mathbb{R}^n \rightarrow \mathbb{R}$ are **equality constraints**
- $\mathcal{X} \subseteq \mathbb{R}^n$ is the **domain** (often $\mathcal{X} = \mathbb{R}^n$)

The **feasible set** is $S = \{\mathbf{x} \in \mathcal{X} : g_i(\mathbf{x}) \leq 0, h_j(\mathbf{x}) = 0\}$.

Convex Optimization: Special Structure

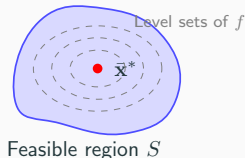
Definition (Convex Optimization Problem) [Course Notes:

Definition]

Problem (1) is a **convex optimization problem** if:

- f is convex
- g_i are convex for all $i = 1, \dots, m$
- h_j are affine: $h_j(\mathbf{x}) = \mathbf{a}_j^T \mathbf{x} - b_j$ for all $j = 1, \dots, p$
- $\mathcal{X}$ is a convex set

Key Property: For convex problems, the feasible set S is convex.



Exercise: Identifying Convex Problems

Quick Check

Determine whether each problem is convex. If not, identify which condition fails.

1. $\min x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \leq 1, \quad x_1 \geq 0$

2. $\min x_1 x_2 \quad \text{s.t.} \quad x_1^2 + x_2^2 \leq 1$

3. $\min e^{x_1} + x_2^2 \quad \text{s.t.} \quad x_1^2 + x_2^2 = 1$

4. $\min \|A\mathbf{x} - \mathbf{b}\|_2^2 \quad \text{s.t.} \quad \mathbf{x} \geq \mathbf{0}$

Hint: Check convexity of objective, constraints, and whether equality constraints are affine.

Solution: Identifying Convex Problems

Answers

1. **Convex.** $f(x) = x_1^2 + x_2^2$ is convex, $g_1(x) = x_1 + x_2 - 1$ and $g_2(x) = -x_1$ are affine (hence convex).
2. **Not convex.** $f(x) = x_1x_2$ is neither convex nor concave (indefinite Hessian). The constraint is convex.
3. **Not convex.** The objective $e^{x_1} + x_2^2$ is convex, but the equality constraint $x_1^2 + x_2^2 = 1$ is not affine.
4. **Convex.** $\|A\mathbf{x} - \mathbf{b}\|_2^2$ is a convex quadratic. Constraints $-x_i \leq 0$ are affine.

The Constrained Optimization Problem

The Lagrangian Function

KKT Conditions

KKT Necessary Conditions

KKT Sufficient Conditions

Constraint Qualifications

Review: Calculus Tools for KKT Proofs

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Applications and Examples

The Lagrangian Function

Definition 15.1 (Lagrangian)

For the constrained problem (1), the **Lagrangian function**

$\mathcal{L} : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$ is:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x})$$

- $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_m) \in \mathbb{R}^m$ are **Lagrange multipliers** for inequality constraints
- $\boldsymbol{\nu} = (\nu_1, \dots, \nu_p) \in \mathbb{R}^p$ are **Lagrange multipliers** for equality constraints
- The multipliers are also called **dual variables**

Key Idea: The Lagrangian “absorbs” the constraints into the objective by penalizing constraint violations.

Why the Lagrangian? An Adversarial Game

Idea: Instead of enforcing constraints directly, set up a **game** between two players.

Claim (No duality needed!)

The constrained problem is **equivalent** to:

$$\min_{\mathbf{x}} \max_{\lambda \geq 0, \nu} \mathcal{L}(\mathbf{x}, \lambda, \nu)$$

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Why? Fix any $\mathbf{x}$ and consider the inner max:

- If $\mathbf{x}$ is **infeasible** (some $g_i(\mathbf{x}) > 0$): the adversary sends $\lambda_i \rightarrow \infty$, so $\mathcal{L} \rightarrow +\infty$.
- If $\mathbf{x}$ is **feasible** ($g_i(\mathbf{x}) \leq 0$, $h_j(\mathbf{x}) = 0$):
 $\lambda_i g_i(\mathbf{x}) \leq 0$ for $\lambda_i \geq 0$, and $\nu_j h_j(\mathbf{x}) = 0$.
Adversary's best: set all $\lambda_i = 0$, giving $\mathcal{L} = f(\mathbf{x})$.

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So the inner max acts as a **constraint enforcer**:

$$\max_{\lambda \geq 0, \nu} \mathcal{L}(\mathbf{x}, \lambda, \nu) = \begin{cases} f(\mathbf{x}) & \text{if } \mathbf{x} \text{ is feasible} \\ +\infty & \text{if } \mathbf{x} \text{ is infeasible} \end{cases}$$

The constraints aren't gone—they're enforced by the adversary's optimization!

From Game Equilibrium to KKT Conditions

At the **equilibrium** of the game $\min_{\mathbf{x}} \max_{\lambda \geq 0, \nu} \mathcal{L}(\mathbf{x}, \lambda, \nu)$, neither player can improve:

Game condition

Minimizer's optimality: $\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0}$

Adversary's constraint: $\lambda \geq \mathbf{0}$

Adversary's optimality: $\lambda_i g_i(\bar{\mathbf{x}}) = 0$

Original feasibility: $g_i(\bar{\mathbf{x}}) \leq 0, h_j(\bar{\mathbf{x}}) = 0$

KKT condition

Stationarity

Dual feasibility

Complementary slackness

Primal feasibility

From Game Equilibrium to KKT Conditions

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Game condition	KKT condition
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Adversary's optimality: $\lambda_i g_i(\bar{\mathbf{x}}) = 0$	Complementary slackness
Original feasibility: $g_i(\bar{\mathbf{x}}) \leq 0, h_j(\bar{\mathbf{x}}) = 0$	Primal feasibility

Why complementary slackness? Think like the adversary (maximizer):

- If $g_i(\bar{\mathbf{x}}) < 0$ (slack): increasing λ_i *decreases* $\mathcal{L} \Rightarrow$ adversary sets $\lambda_i = 0$.
- If $g_i(\bar{\mathbf{x}}) = 0$ (active): the $\lambda_i g_i$ term vanishes regardless $\Rightarrow \lambda_i$ can be positive.

From Game Equilibrium to KKT Conditions

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- If $g_i(\bar{\mathbf{x}}) = 0$ (active): the $\lambda_i g_i$ term vanishes regardless $\Rightarrow \lambda_i$ can be positive.

Either way: $\lambda_i g_i(\bar{\mathbf{x}}) = 0$. **The KKT conditions are not magic—they are the **equilibrium conditions** of the Lagrangian game.**

Why the Lagrangian? Bounding the Optimal Value

Consider the primal problem:

$$p^* = \min_{\mathbf{x}} f(\mathbf{x}) \quad \text{s.t.} \quad g_i(\mathbf{x}) \leq 0, h_j(\mathbf{x}) = 0$$

Key Observation

If $\mathbf{x}$ is **feasible**, then $g_i(\mathbf{x}) \leq 0$ and $h_j(\mathbf{x}) = 0$. For any $\lambda \geq 0$ and any $\nu \in \mathbb{R}^p$:

$$\mathcal{L}(\mathbf{x}, \lambda, \nu) = f(\mathbf{x}) + \sum_i \lambda_i \underbrace{g_i(\mathbf{x})}_{\leq 0} + \sum_j \nu_j \underbrace{h_j(\mathbf{x})}_{=0} \leq f(\mathbf{x})$$

Lower Bounding Property

Taking infimum over all $\mathbf{x}$:

$$\underbrace{\inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda, \nu)}_{\text{dual function } g(\lambda, \nu)} \leq p^*$$

For all $\lambda \geq 0$ and all ν : $g(\lambda, \nu) \leq p^*$

Interpretation: Each choice of multipliers (λ, ν) yields a lower bound on p^* . The best bound comes from the *dual problem* (Lecture 7).

Lagrangian Bounds: Warm-Up Example

Problem

$$\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad x + y \geq 1 \quad \Leftrightarrow \quad 1 - x - y \leq 0$$

Lagrangian and Dual Function

$\mathcal{L}(x, y, \lambda) = x^2 + y^2 + \lambda(1 - x - y)$ for $\lambda \geq 0$.

Minimizing over (x, y) : $\nabla_{x,y} \mathcal{L} = \mathbf{0}$ gives $x = y = \lambda/2$, so:

$$g(\lambda) = \left(\frac{\lambda}{2}\right)^2 + \left(\frac{\lambda}{2}\right)^2 + \lambda(1 - \lambda) = \lambda - \frac{\lambda^2}{2}$$

Lower Bounds from Different Multipliers

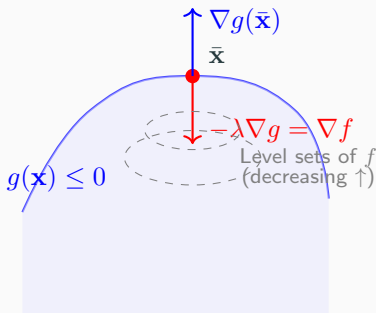
- $\lambda = 0$: $g(0) = 0 \leq p^* = 1/2$ ✓ (trivial)
- $\lambda = 0.5$: $g(0.5) = 0.375 \leq 1/2$ ✓ (better!)
- $\lambda = 1$: $g(1) = 0.5 = p^*$ ✓ (tight! $\lambda^* = 1$)
- $\lambda = 1.5$: $g(1.5) = 0.375 \leq 1/2$ ✓ (overshooting)

Takeaway: The optimal multiplier λ^* gives **zero duality gap**: $g(\lambda^*) = p^*$.

This is *strong duality*—more in Lecture 7.

Geometric Interpretation of Lagrange Multipliers

For a constrained minimum, the gradient of f must be a linear combination of constraint gradients.



At an optimal point: $\nabla f(\bar{\mathbf{x}}) + \lambda \nabla g(\bar{\mathbf{x}}) = \mathbf{0}$, so $\nabla f = -\lambda \nabla g$ with $\lambda \geq 0$.

Exercise: Writing the Lagrangian

Practice

Write the Lagrangian for each problem:

1. $\min x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 = 1$

2. $\min x_1 + 2x_2 \quad \text{s.t.} \quad x_1^2 + x_2^2 \leq 4, \quad x_1 \geq 0$

3. $\min c^T \mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}$

Hint: Use λ_i for inequality constraints (written as $g_i(\mathbf{x}) \leq 0$) and ν_j for equality constraints.

Solution: Writing the Lagrangian

Answers

1. $\mathcal{L}(x_1, x_2, \nu) = x_1^2 + x_2^2 + \nu(x_1 + x_2 - 1)$
2. First rewrite: $g_1(\mathbf{x}) = x_1^2 + x_2^2 - 4$, $g_2(\mathbf{x}) = -x_1$.

$$\mathcal{L}(x_1, x_2, \lambda_1, \lambda_2) = x_1 + 2x_2 + \lambda_1(x_1^2 + x_2^2 - 4) + \lambda_2(-x_1)$$

3. Constraints: $A\mathbf{x} - \mathbf{b} = \mathbf{0}$ and $-\mathbf{x} \leq \mathbf{0}$.

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = c^T \mathbf{x} + \boldsymbol{\nu}^T (A\mathbf{x} - \mathbf{b}) + \boldsymbol{\lambda}^T (-\mathbf{x})$$

$$\text{or equivalently: } \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = c^T \mathbf{x} + \boldsymbol{\nu}^T (A\mathbf{x} - \mathbf{b}) - \boldsymbol{\lambda}^T \mathbf{x}$$

Roadmap

The Constrained Optimization Problem

The Lagrangian Function

KKT Conditions

KKT Necessary Conditions

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Constraint Qualifications

Review: Calculus Tools for KKT Proofs

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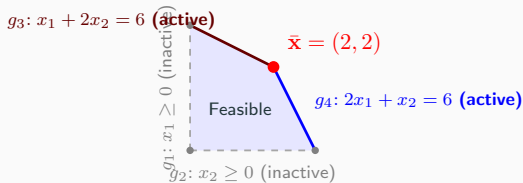
Applications and Examples

Active and Inactive Constraints

Definition 15.4 (Active/Binding Constraints)

At a feasible point $\bar{\mathbf{x}}$:

- Constraint $g_i(\mathbf{x}) \leq 0$ is **active** (or binding) if $g_i(\bar{\mathbf{x}}) = 0$
- Constraint $g_i(\mathbf{x}) \leq 0$ is **inactive** if $g_i(\bar{\mathbf{x}}) < 0$
- Equality constraints $h_j(\mathbf{x}) = 0$ are always active



At $\bar{\mathbf{x}} = (2, 2)$: g_1, g_2 inactive (slack > 0), g_3, g_4 active (binding). Active set: $\mathcal{A}(\bar{\mathbf{x}}) = \{3, 4\}$.

The KKT Conditions

Karush-Kuhn-Tucker (KKT) Conditions [Theorem 15.7 / Definition

15.8-15.9 in Course Notes]

For problem (1), the KKT conditions at $\bar{\mathbf{x}}$ are:

1. **Stationarity:** $\nabla f(\bar{\mathbf{x}}) + \sum_{i=1}^m \lambda_i \nabla g_i(\bar{\mathbf{x}}) + \sum_{j=1}^p \nu_j \nabla h_j(\bar{\mathbf{x}}) = \mathbf{0}$
2. **Primal Feasibility:** $g_i(\bar{\mathbf{x}}) \leq 0$ for all i ; $h_j(\bar{\mathbf{x}}) = 0$ for all j
3. **Dual Feasibility:** $\lambda_i \geq 0$ for all i
4. **Complementary Slackness:** $\lambda_i g_i(\bar{\mathbf{x}}) = 0$ for all i

Interpretation:

- Stationarity: $\nabla_{\mathbf{x}} \mathcal{L}(\bar{\mathbf{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = \mathbf{0}$
- Complementary slackness: either $\lambda_i = 0$ or $g_i(\bar{\mathbf{x}}) = 0$ (or both)

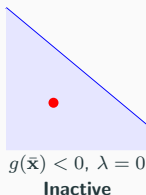
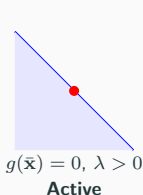
Understanding Complementary Slackness

Complementary Slackness: $\lambda_i g_i(\bar{\mathbf{x}}) = 0$

For each inequality constraint, one of the following holds:

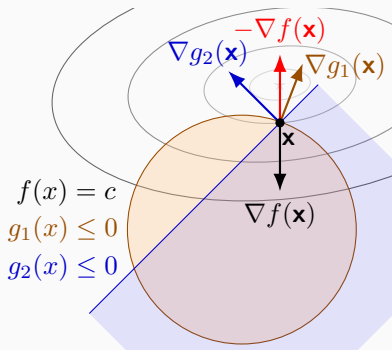
- $g_i(\bar{\mathbf{x}}) = 0$ (constraint is active) — multiplier λ_i can be positive
- $g_i(\bar{\mathbf{x}}) < 0$ (constraint is inactive) — multiplier λ_i must be zero

Intuition: Inactive constraints don't affect the optimum, so their multipliers are zero. Only active constraints “push” on the solution.



Geometric Interpretation: KKT Optimal Point

At an optimal point, the **negative gradient** of the objective must lie in the cone generated by the active constraint gradients.



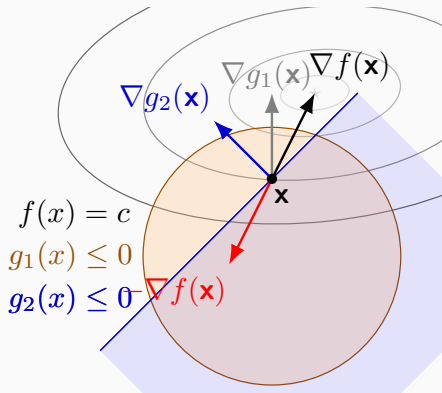
Key insight: $-\nabla f(\bar{\mathbf{x}})$ lies in the **cone** of active constraint gradients:

$$-\nabla f(\bar{\mathbf{x}}) = \sum_{i \in \mathcal{A}} \lambda_i \nabla g_i(\bar{\mathbf{x}})$$

with $\lambda_i \geq 0$ (dual feasibility).

Geometric Interpretation: Non-Optimal Point

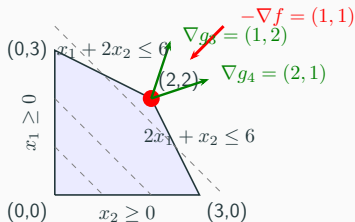
At a non-optimal point, the **negative gradient** of the objective is *not* in the cone of active constraint gradients.



Observation: When KKT conditions fail, a feasible direction of descent exists, so the point cannot be optimal.

KKT Geometry: Linear Objective over a Polytope

For **linear** objectives over polytopes, optima occur at vertices. At the vertex where multiple constraints are active, the KKT stationarity condition determines the multipliers uniquely.



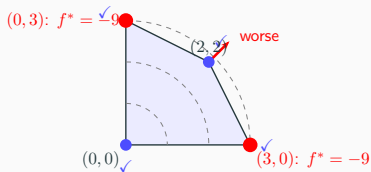
At the optimal vertex $(2,2)$, both constraints are active. The KKT stationarity condition uniquely determines:

$$-\nabla f = \lambda_3 \nabla g_3 + \lambda_4 \nabla g_4 \implies (1, 1) = \lambda_3 (1, 2) + \lambda_4 (2, 1)$$

Solving: $\lambda_3 = \lambda_4 = 1/3 > 0$ (dual feasible). This is the only KKT point.

KKT Geometry: Concave Objective over a Polytope

Caution: For concave (non-convex) minimization over polytopes, KKT conditions can be satisfied at **every vertex**! KKT is necessary but not sufficient here.



Objective: $\min -(x_1^2 + x_2^2)$ subject to polytope.

The level curves are circles centered at the origin. Since we minimize a concave function, the optimum is at a vertex **farthest from the origin**: $(3,0)$ or $(0,3)$ with objective value -9 .

Key difference: KKT conditions hold at **all four vertices**, but only $(3,0)$ and $(0,3)$ are optimal.

Why convexity matters: For convex problems, any KKT point is globally optimal. Without convexity, you must check all KKT candidates.

Exercise: Writing KKT Conditions

Practice

Write the KKT conditions for:

$$\min x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \geq 1$$

First, rewrite the constraint in standard form ($g(\mathbf{x}) \leq 0$), then write:

1. Stationarity condition:
2. Primal feasibility:
3. Dual feasibility:
4. Complementary slackness:

Solution: Writing KKT Conditions

Answers

Rewrite: $g(\mathbf{x}) = 1 - x_1 - x_2 \leq 0$

Lagrangian: $\mathcal{L}(x_1, x_2, \lambda) = x_1^2 + x_2^2 + \lambda(1 - x_1 - x_2)$

1. Stationarity:

$$\frac{\partial \mathcal{L}}{\partial x_1} = 2x_1 - \lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 2x_2 - \lambda = 0$$

2. Primal feasibility: $1 - x_1 - x_2 \leq 0$ (i.e., $x_1 + x_2 \geq 1$)

3. Dual feasibility: $\lambda \geq 0$

4. Complementary slackness: $\lambda(1 - x_1 - x_2) = 0$

KKT Conditions: Necessary for Optimality

Theorem 15.7 (KKT Necessary Conditions)

Let $\bar{x}$ be a local minimum of (1) where f , g_i , and h_j are continuously differentiable. If a **constraint qualification** (e.g., LICQ) holds at $\bar{x}$, then there exist multipliers $\lambda^* \in \mathbb{R}^m$ and $\nu^* \in \mathbb{R}^p$ such that $(\bar{x}, \lambda^*, \nu^*)$ satisfies the KKT conditions.

Key Points:

- KKT conditions are necessary: optimal $\Rightarrow$ KKT point
- But not every KKT point is optimal (in general)
- Need constraint qualification to ensure multipliers exist

Proof: KKT Necessary Conditions

Theorem (restated)

If $\bar{\mathbf{x}}$ is a local minimum and a constraint qualification holds, then there exist KKT multipliers $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$.

Proof Outline: (to be done on board)

1. Define the active constraint set $\mathcal{A} = \{i : g_i(\bar{\mathbf{x}}) = 0\}$
2. At a local minimum, no feasible descent direction exists
3. Use Fritz John Necessary Conditions (Theorem 15.6 in Course Notes) to obtain initial multipliers
4. Constraint qualification ensures $\lambda_0 \neq 0$, so we can normalize to $\lambda_0 = 1$
5. This gives the KKT conditions with proper multipliers

See instructor proof notes for complete details.

KKT Conditions: Sufficient for Convex Problems

Theorem 15.11 (KKT Sufficient Conditions for Convex Optimization)

Consider a **convex** optimization problem (i.e., f and g_i are convex, h_j are affine). If $(\bar{\mathbf{x}}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ satisfies the KKT conditions, then $\bar{\mathbf{x}}$ is a **global optimal solution**.

This is why convex optimization is special!

- For convex problems: KKT conditions are both necessary and sufficient
- Finding a KKT point = finding the global optimum
- No need to check second-order conditions

The KKT Philosophy for Convex Problems

$\bar{\mathbf{x}}$ is optimal $\iff (\bar{\mathbf{x}}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ satisfies KKT

Proof: KKT Sufficient Conditions

Theorem (restated)

For convex problems, KKT conditions imply global optimality.

Proof Outline: (to be done on board)

1. Suppose $(\bar{\mathbf{x}}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ satisfies KKT
2. For any feasible $\mathbf{x}$, use convexity of f :

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}})$$

3. Use stationarity: $\nabla f(\bar{\mathbf{x}}) = -\sum_i \lambda_i^* \nabla g_i(\bar{\mathbf{x}}) - \sum_j \nu_j^* \nabla h_j(\bar{\mathbf{x}})$
4. Use dual feasibility, complementary slackness, and feasibility of $\mathbf{x}$ to show $f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$

See instructor proof notes for complete details.

Exercise: Solving via KKT Conditions

Group Exercise

Solve using KKT conditions:

$$\min x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \geq 1$$

1. From stationarity, what is the relationship between x_1 , x_2 , and λ ?
2. Using complementary slackness, consider two cases: $\lambda = 0$ or $x_1 + x_2 = 1$.
3. Find the optimal solution $(\bar{x}_1, \bar{x}_2)$ and multiplier λ^* .

Hint: If $\lambda = 0$, what does stationarity tell you?

Solution: Solving via KKT Conditions

Answers

1. From stationarity: $2x_1 = \lambda$ and $2x_2 = \lambda$. Thus $x_1 = x_2 = \lambda/2$.
2. **Case 1:** $\lambda = 0$. Then $x_1 = x_2 = 0$, but this violates primal feasibility ($0 + 0 \not\geq 1$). Infeasible.
Case 2: $x_1 + x_2 = 1$. From (1): $\lambda/2 + \lambda/2 = 1$, so $\lambda = 1$.
3. From $\lambda = 1$: $x_1 = x_2 = 1/2$.

Optimal solution: $(\bar{x}_1, \bar{x}_2) = (1/2, 1/2)$ with $\lambda^* = 1$.

Verification: Check all KKT conditions:

- Stationarity: $2(1/2) - 1 = 0 \checkmark$
- Primal feasibility: $1/2 + 1/2 = 1 \geq 1 \checkmark$
- Dual feasibility: $\lambda = 1 \geq 0 \checkmark$
- Complementary slackness: $1 \cdot (1 - 1) = 0 \checkmark$

Worked Example: KKT with Multiple Constraints

Find the Minimum

$$\min (x_1 - 2)^2 + (x_2 - 2)^2 \quad \text{s.t.} \quad x_1 + x_2 \leq 3, \quad x_1 \geq 0, \quad x_2 \geq 0$$

1. Write the KKT conditions (4 conditions).
2. Use complementary slackness to enumerate possible cases.
3. Solve each case and check feasibility.

Hint: There are 3 constraints, so complementary slackness gives $2^3 = 8$ possible cases (most are quickly ruled out).

Solution: KKT with Multiple Constraints

Answers

KKT Conditions: Rewrite constraints as $g_1(x) = x_1 + x_2 - 3 \leq 0$,
 $g_2(x) = -x_1 \leq 0$, $g_3(x) = -x_2 \leq 0$.

- **Stationarity:** $2(x_1 - 2) + \lambda_1 - \lambda_2 = 0$ and $2(x_2 - 2) + \lambda_1 - \lambda_3 = 0$
- **Primal feasibility:** $x_1 + x_2 \leq 3$, $x_1 \geq 0$, $x_2 \geq 0$
- **Dual feasibility:** $\lambda_1, \lambda_2, \lambda_3 \geq 0$
- **Comp. slackness:** $\lambda_1(x_1 + x_2 - 3) = 0$, $\lambda_2(-x_1) = 0$, $\lambda_3(-x_2) = 0$

Case Analysis:

Case A: $\lambda_1 = 0$ (first constraint inactive). Stationarity $\Rightarrow$
 $x_1 = 2 + \lambda_2/2$, $x_2 = 2 + \lambda_3/2$. Then $x_1 + x_2 = 4 + (\lambda_2 + \lambda_3)/2 > 3$
(since $\lambda_i \geq 0$). **Infeasible!**

Case B: $x_1 + x_2 = 3$, $\lambda_2 = \lambda_3 = 0$ (first active, others inactive).
Stationarity: $2(x_1 - 2) + \lambda_1 = 0$ and $2(x_2 - 2) + \lambda_1 = 0$. Both give
 $\lambda_1 = 4 - 2x_1 = 4 - 2x_2$, so $x_1 = x_2 = 3/2$. Then $\lambda_1 = 1 > 0$ ✓, and
 $x_1, x_2 > 0$ ✓. **Feasible!**

Case C: $x_1 + x_2 = 3$, $x_1 = 0$. Then $x_2 = 3$. Stationarity:
 $2(0 - 2) + \lambda_1 - \lambda_2 = 0 \Rightarrow \lambda_1 - \lambda_2 = -4$. Also
 $2(3 - 2) + \lambda_1 = 0 \Rightarrow \lambda_1 = -2 < 0$. **Violates dual feasibility!**

Case D: $x_1 + x_2 = 3$, $x_2 = 0$. By symmetry with Case C. **Also fails.**

Conclusion: Unique KKT point is $\bar{x} = (3/2, 3/2)$ with $\lambda_1^* = 1$,
 $\lambda_2^* = \lambda_3^* = 0$. Optimal value $f^* = (3/2 - 2)^2 + (3/2 - 2)^2 = 1/2$.
Since convex, this is **globally optimal**.

The Constrained Optimization Problem

The Lagrangian Function

KKT Conditions

KKT Necessary Conditions

KKT Sufficient Conditions

Constraint Qualifications

Review: Calculus Tools for KKT Proofs

Proof Space: Key Results

Applications and Examples

Why Constraint Qualifications?

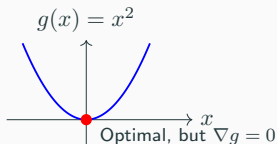
Problem: KKT Multipliers May Not Exist

Even at an optimal point, KKT conditions may fail without a constraint qualification!

Example: $\min x \quad \text{s.t.} \quad x^2 \leq 0$

The only feasible point is $x = 0$, which is optimal.

KKT stationarity requires: $1 + \lambda \cdot 2x = 0$. At $x = 0$: $1 + 0 = 1 \neq 0$. **No**
 λ works!



Problem: At $x = 0$, $\nabla g(0) = 0$. The constraint gradient "vanishes."

Linear Independence Constraint Qualification (LICQ)

Definition 15.7 (Linear Independence Constraint Qualification — LICQ)

The **Linear Independence Constraint Qualification** holds at $\bar{\mathbf{x}}$ if the gradients of all active constraints are **linearly independent**:

$$\{\nabla g_i(\bar{\mathbf{x}}) : i \in \mathcal{A}\} \cup \{\nabla h_j(\bar{\mathbf{x}}) : j = 1, \dots, p\}$$

where $\mathcal{A} = \{i : g_i(\bar{\mathbf{x}}) = 0\}$ is the active set.

Implications:

- LICQ $\Rightarrow$ KKT multipliers exist and are unique
- Easy to check for linear/affine constraints
- May fail at “pathological” points

Slater's Condition (for Convex Problems)

Definition 15.11 (Slater's Condition)

For a convex problem, Slater's condition holds if there exists a strictly feasible point $\tilde{\mathbf{x}}$:

- $g_i(\tilde{\mathbf{x}}) < 0$ for all $i = 1, \dots, m$ (strict inequality)
- $h_j(\tilde{\mathbf{x}}) = 0$ for all $j = 1, \dots, p$

Key Properties:

- Slater's condition is very mild—usually satisfied in practice
- For linear inequalities, can relax to $g_i(\tilde{\mathbf{x}}) \leq 0$
- Guarantees KKT multipliers exist at any optimal point

For Convex Problems with Slater's Condition

Optimal $\iff$ KKT conditions hold

Exercise: Checking Constraint Qualifications

Quick Check

For each problem, determine if Slater's condition holds.

1. $\min x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \leq 1, \quad x_1 - x_2 \leq 1$

2. $\min x_1 + x_2 \quad \text{s.t.} \quad x_1^2 + x_2^2 \leq 1$

3. $\min x_1 \quad \text{s.t.} \quad (x_1 - 1)^2 + x_2^2 \leq 0$

*Hint: Can you find a point that satisfies all inequalities **strictly**?*

Solution: Checking Constraint Qualifications

Answers

1. **Slater holds.** Take $\tilde{\mathbf{x}} = (0, 0)$: $0 + 0 = 0 < 1$ and $0 - 0 = 0 < 1$. Both constraints strictly satisfied.
2. **Slater holds.** Take $\tilde{\mathbf{x}} = (0, 0)$: $0^2 + 0^2 = 0 < 1$. Strictly inside the unit disk.
3. **Slater fails.** The constraint $(x_1 - 1)^2 + x_2^2 \leq 0$ requires $(x_1 - 1)^2 + x_2^2 = 0$ (sum of squares). The only feasible point is $(1, 0)$, which is on the boundary, not strictly feasible.

Note: Problem (3) is the pathological example where $g(\mathbf{x}) = (x_1 - 1)^2 + x_2^2$ is convex but the feasible region is a single point!

Fritz John Necessary Conditions

Theorem (Fritz John Conditions)

If $\bar{\mathbf{x}}$ is a local minimizer of $\min f(\mathbf{x})$ s.t. $g_i(\mathbf{x}) \leq 0$, then there exist multipliers $\lambda_0 \geq 0$ and $\lambda_i \geq 0$ ($i = 1, \dots, m$), **not all zero**, such that:

1. $\lambda_0 \nabla f(\bar{\mathbf{x}}) + \sum_{i=1}^m \lambda_i \nabla g_i(\bar{\mathbf{x}}) = 0$
2. $\lambda_i g_i(\bar{\mathbf{x}}) = 0$ for all i (complementary slackness)
3. $(\lambda_0, \lambda_1, \dots, \lambda_m) \neq \mathbf{0}$

Key insight: Fritz John holds at *any* local minimizer with *no* assumptions on the constraints.

Relationship to KKT:

- KKT = Fritz John with $\lambda_0 > 0$ (divide through by λ_0 and normalize to $\lambda_0 = 1$)
- If $\lambda_0 = 0$, the objective plays no role — the conditions only reflect constraint geometry
- A **constraint qualification** guarantees $\lambda_0 > 0$, giving KKT

Mangasarian–Fromovitz CQ (MFCQ)

Definition (MFCQ)

The **Mangasarian–Fromovitz Constraint Qualification** holds at $\bar{\mathbf{x}}$ if there exists a direction $d \in \mathbb{R}^n$ such that:

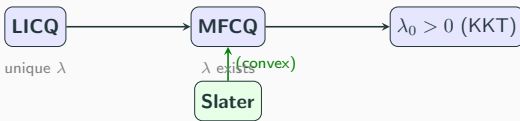
$$\nabla g_i(\bar{\mathbf{x}})^T d < 0 \quad \text{for all } i \in \mathcal{A}(\bar{\mathbf{x}})$$

where $\mathcal{A}(\bar{\mathbf{x}}) = \{i : g_i(\bar{\mathbf{x}}) = 0\}$ is the active set.

(For problems with equality constraints $h_j(\mathbf{x}) = 0$, also require $\{\nabla h_j(\bar{\mathbf{x}})\}$ linearly independent and $\nabla h_j(\bar{\mathbf{x}})^T d = 0$ for all j .)

Interpretation: There exists a single direction d that simultaneously moves *strictly into the interior* of every active inequality constraint.

Constraint Qualification Hierarchy:



Exercise: When KKT Breaks

Group Exercise (8 min)

Consider the problem:

$$\min -x_1 \quad \text{s.t.} \quad g_1 : x_2 - (1 - x_1)^3 \leq 0, \quad g_2 : -x_2 \leq 0$$

The feasible set is $\{(\mathbf{x}) : 0 \leq x_2 \leq (1 - x_1)^3, x_1 \leq 1\}$.

1. Find $\mathbf{x}^*$ (the optimal solution).
2. Compute $\nabla g_1(\mathbf{x}^*)$ and $\nabla g_2(\mathbf{x}^*)$.
3. Try to write KKT stationarity at $\mathbf{x}^*$. What goes wrong?
4. Write the Fritz John conditions and find valid $(\lambda_0, \lambda_1, \lambda_2)$. What is λ_0 ?

Hint: Both constraints are active at $\mathbf{x}^$. Look at the first component of stationarity.*

Solution: When KKT Breaks

Answers

(1) $\mathbf{x}^* = (1, 0)$, $f(\mathbf{x}^*) = -1$. Both constraints are active.

(2) $g_1(\mathbf{x}) = x_2 - (1 - x_1)^3$, so $\nabla g_1 = (3(1 - x_1)^2, 1)^T$. At $\mathbf{x}^*$:
 $\nabla g_1(1, 0) = (0, 1)^T$.

$\nabla g_2 = (0, -1)^T$ everywhere.

(3) KKT stationarity: $\begin{pmatrix} -1 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Component 1: $-1 = 0$. **Contradiction!** No KKT multipliers exist.

(4) Fritz John: $-\lambda_0 = 0 \Rightarrow \lambda_0 = 0$. Then $\lambda_1 = \lambda_2 = 1$ works.

The objective dropped out entirely because $\lambda_0 = 0$. The active constraint gradients $(0, 1)^T$ and $(0, -1)^T$ are positively dependent — they point in opposite directions, violating both LICQ and MFCQ.

Exercise: Diagnosing CQ Failures

Group Exercise (8 min)

Consider: $\min x_1 \quad \text{s.t.} \quad g_1 : -x_1 \leq 0, \quad g_2 : -2x_1 \leq 0.$

1. Find $\mathbf{x}^*$. Which constraints are active?
2. Check LICQ at $\mathbf{x}^*$: are the active constraint gradients linearly independent?
3. Check MFCQ at $\mathbf{x}^*$: find a direction d with $\nabla g_i(\mathbf{x}^*)^T d < 0$ for all active i .
4. Find **all** KKT multiplier vectors (λ_1, λ_2) . How many solutions are there?

Compare with the previous exercise: LICQ fails in both, but MFCQ behaves differently!

Solution: Diagnosing CQ Failures

Answers

(1) $\mathbf{x}^* = (0, 0)$. Both constraints are active.

(2) $\nabla g_1 = (-1, 0)^T$, $\nabla g_2 = (-2, 0)^T$. These are parallel (linearly dependent). **LICQ fails.**

(3) Try $d = (1, 0)$: $\nabla g_1^T d = -1 < 0$ and $\nabla g_2^T d = -2 < 0$. **MFCQ holds!**

(4) KKT stationarity: $(1, 0)^T + \lambda_1(-1, 0)^T + \lambda_2(-2, 0)^T = 0$ gives $\lambda_1 + 2\lambda_2 = 1$.

All solutions: $(\lambda_1, \lambda_2) = (1 - 2t, t)$ for $t \in [0, 1/2]$. **Infinitely many!**

Three levels of CQ behavior:

	LICQ	MFCQ	KKT
Previous exercise	×	×	Don't exist
This exercise	×	✓	Exist, not unique
LICQ holds	✓	✓	Exist & unique

Exercise: Verifying Constraint Qualifications

Practice

For each problem, determine which CQs hold at the given point $\bar{\mathbf{x}}$. Justify each.

1. $g_1(\mathbf{x}) = x_1^2 + x_2^2 - 1 \leq 0, \quad \bar{\mathbf{x}} = (1, 0).$

2. $g_1(\mathbf{x}) = x_1 \leq 0, \quad g_2(\mathbf{x}) = -x_1 \leq 0, \quad \bar{\mathbf{x}} = (0, 0).$

3. $g_1(\mathbf{x}) = x_1^3 \leq 0, \quad \bar{\mathbf{x}} = (0, 0).$

4. $g_1(\mathbf{x}) = -x_1 \leq 0, \quad g_2(\mathbf{x}) = -x_2 \leq 0, \quad h_1(\mathbf{x}) = x_1 + x_2 - 1 = 0,$
 $\bar{\mathbf{x}} = (1, 0).$

Check: (i) which constraints are active, (ii) LICQ, (iii) MFCQ, (iv) Slater's (if applicable).

Solution: Verifying Constraint Qualifications

Answers

(1) One active constraint. $\nabla g_1(1,0) = (2,0)^T \neq 0$. LICQ ✓, MFCQ ✓ ($d = (-1,0)$), Slater ✓ ($\hat{x} = (0,0)$ gives $g_1 = -1 < 0$).

(2) Both active. $\nabla g_1 = (1,0)^T$, $\nabla g_2 = (-1,0)^T$: dependent. LICQ ✗. MFCQ needs $d_1 < 0$ and $d_1 > 0$ simultaneously — impossible. MFCQ ✗. Slater needs $\hat{x}_1 < 0$ and $-\hat{x}_1 < 0$ — impossible. Slater ✗.

(3) Active: $g_1(0,0) = 0$. But $\nabla g_1 = (3x_1^2, 0)^T|_{(0,0)} = (0,0)^T$ — the gradient **vanishes!** LICQ ✗, MFCQ ✗ (need $(0,0)^T d < 0$, impossible). Not convex, so Slater does not apply.

(4) g_1 inactive ($g_1(1,0) = -1$), g_2 active, h_1 active. Active gradients: $\nabla g_2 = (0,-1)^T$, $\nabla h_1 = (1,1)^T$: linearly independent. LICQ ✓. MFCQ: $d = (1,-1)$ gives $\nabla g_2^T d = 1 > 0$... need $d_2 > 0$ from g_2 , and $d_1 + d_2 = 0$. Try $d = (-1,1)$: $\nabla g_2^T d = -1 \cdot 0 + 1 \cdot (-1)$...

Actually let me fix this: $g_2 = -x_2$, so $\nabla g_2 = (0,-1)^T$. Need $(0,-1)^T d < 0$, i.e., $d_2 > 0$. Need $(1,1)^T d = 0$, i.e., $d_1 = -d_2$. Take $d = (-1,1)$. MFCQ ✓.

Exercise: Proving MFCQ $\Rightarrow$ KKT

Proof Exercise

Suppose $\bar{\mathbf{x}}$ is a local minimizer and MFCQ holds: $\exists d$ with $\nabla g_i(\bar{\mathbf{x}})^T d < 0$ for all $i \in \mathcal{A}(\bar{\mathbf{x}})$. Fritz John gives $\lambda_0 \geq 0$, $\lambda_i \geq 0$ (not all zero) with $\lambda_0 \nabla f(\bar{\mathbf{x}}) + \sum_{i \in \mathcal{A}} \lambda_i \nabla g_i(\bar{\mathbf{x}}) = 0$.

1. Suppose for contradiction that $\lambda_0 = 0$. What does Fritz John reduce to?
2. Take the inner product of the Fritz John equation with the MFCQ direction d . Show this leads to a contradiction.
3. Conclude: $\lambda_0 > 0$, and therefore KKT conditions hold.

Hint: Each term $\lambda_i \nabla g_i(\bar{\mathbf{x}})^T d$ is (non-negative) $\times$ (strictly negative).

Solution: Proving MFCQ $\Rightarrow$ KKT

Proof

(1) With $\lambda_0 = 0$: $\sum_{i \in \mathcal{A}} \lambda_i \nabla g_i(\bar{\mathbf{x}}) = 0$ with $\lambda_i \geq 0$, at least one $\lambda_i > 0$.

(2) Take inner product with d :

$$\sum_{i \in \mathcal{A}} \lambda_i \underbrace{\nabla g_i(\bar{\mathbf{x}})^T d}_{< 0} = 0$$

Each term: $\underbrace{\lambda_i}_{\geq 0} \times \underbrace{\nabla g_i(\bar{\mathbf{x}})^T d}_{< 0} \leq 0$, with at least one term strictly < 0 .

A sum of non-positive terms, at least one strictly negative, **cannot be zero**. **Contradiction!**

(3) Since $\lambda_0 = 0$ leads to a contradiction, we must have $\lambda_0 > 0$.

Dividing Fritz John by λ_0 gives the standard KKT conditions. ■

Why this fails without MFCQ: Without a direction d making all active constraint gradients strictly negative, the active gradients can be “positively dependent”: $\sum \lambda_i \nabla g_i = 0$ with $\lambda_i \geq 0$ can hold, and the inner product argument produces no contradiction.

Roadmap

The Constrained Optimization Problem

The Lagrangian Function

KKT Conditions

KKT Necessary Conditions

KKT Sufficient Conditions

Constraint Qualifications

Review: Calculus Tools for KKT Proofs

Proof Space: Key Results

Applications and Examples

Review: The Gradient of a Multivariate Function

Definition: Let $S \subseteq \mathbb{R}^n$ be nonempty. A function $f : S \rightarrow \mathbb{R}$ is said to be **differentiable** at $\bar{\mathbf{x}} \in \text{int}(S)$ if there exists a gradient vector $\nabla f(\bar{\mathbf{x}})$ and a function $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$f(\mathbf{x}) = \underbrace{f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}})}_{\substack{\text{First-order Taylor series} \\ \text{approximation of } f \text{ near } \bar{\mathbf{x}}}} + \|\mathbf{x} - \bar{\mathbf{x}}\| \alpha(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}) \quad \text{for each } \mathbf{x} \in S,$$

where $\lim_{\mathbf{x} \rightarrow \bar{\mathbf{x}}} \alpha(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}) = 0$.

Review: The Gradient of a Multivariate Function

Definition: Let $S \subseteq \mathbb{R}^n$ be nonempty. A function $f : S \rightarrow \mathbb{R}$ is said to be **differentiable** at $\bar{\mathbf{x}} \in \text{int}(S)$ if there exists a **gradient vector** $\nabla f(\bar{\mathbf{x}})$ and a function $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$f(\mathbf{x}) = \underbrace{f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}})}_{\substack{\text{First-order Taylor series} \\ \text{approximation of } f \text{ near } \bar{\mathbf{x}}}} + \|\mathbf{x} - \bar{\mathbf{x}}\| \alpha(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}) \quad \text{for each } \mathbf{x} \in S,$$

where $\lim_{\mathbf{x} \rightarrow \bar{\mathbf{x}}} \alpha(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}) = 0$.

Note: If f is differentiable at $\bar{\mathbf{x}}$, there is a *unique* **gradient vector** given by

$$\nabla f(\bar{\mathbf{x}}) = \left(\frac{\partial f(\bar{\mathbf{x}})}{\partial x_1}, \dots, \frac{\partial f(\bar{\mathbf{x}})}{\partial x_n} \right) \in \mathbb{R}^n$$

This definition is the foundation for the stationarity condition in the KKT proof.

Review: Taylor Series Approximations

Taylor Series: Given a smooth univariate function $f : \mathbb{R} \rightarrow \mathbb{R}$, we can approximate it with a polynomial around $\bar{x} = a$ as follows:

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2!}f''(a)(x - a)^2 + \frac{1}{3!}f'''(a)(x - a)^3 + \dots$$

Review: Taylor Series Approximations

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Taylor Series Approximations for Multivariate Functions: Given a smooth multivariate function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and a point $\bar{\mathbf{x}} \in \mathbb{R}^n$:

- **Linear/First-order approximation:** $f(\mathbf{x}) \approx f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}})$

Review: Taylor Series Approximations

Taylor Series: Given a smooth univariate function $f : \mathbb{R} \rightarrow \mathbb{R}$, we can approximate it with a polynomial around $\bar{x} = a$ as follows:

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2!}f''(a)(x - a)^2 + \frac{1}{3!}f'''(a)(x - a)^3 + \dots$$

Taylor Series Approximations for Multivariate Functions: Given a smooth multivariate function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and a point $\bar{\mathbf{x}} \in \mathbb{R}^n$:

- Linear/First-order approximation: $f(\mathbf{x}) \approx f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}})$
- Second-order approximation:

$$f(\mathbf{x}) \approx f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) + \frac{1}{2}(\mathbf{x} - \bar{\mathbf{x}})^T H(\bar{\mathbf{x}})(\mathbf{x} - \bar{\mathbf{x}})$$

Hessian matrix $H(\bar{\mathbf{x}}) = \nabla^2 f(\bar{\mathbf{x}})$ is defined by $H_{ij}(\bar{\mathbf{x}}) = \frac{\partial^2 f(\bar{\mathbf{x}})}{\partial x_i \partial x_j}$, $\forall i, j \in [n]$

Review: Twice Differentiable Functions

Definition: Let S be a nonempty set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. Then f is said to be twice differentiable at $\bar{\mathbf{x}} \in \text{int}(S)$ if there exists

- a gradient vector $\nabla f(\bar{\mathbf{x}})$,
- an $n \times n$ symmetric Hessian matrix $H(\bar{\mathbf{x}}) = \nabla^2 f(\bar{\mathbf{x}})$, and
- a function $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}$

such that for each $\mathbf{x} \in S$:

$$f(\mathbf{x}) = f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) + \underbrace{\frac{1}{2}(\mathbf{x} - \bar{\mathbf{x}})^T H(\bar{\mathbf{x}})(\mathbf{x} - \bar{\mathbf{x}})}_{\text{Second-order Taylor series approximation of } f \text{ near } \bar{\mathbf{x}}} + \|\mathbf{x} - \bar{\mathbf{x}}\|^2 \alpha(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}).$$

Review: Twice Differentiable Functions

Definition: Let S be a nonempty set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. Then f is said to be twice differentiable at $\bar{\mathbf{x}} \in \text{int}(S)$ if there exists

- a gradient vector $\nabla f(\bar{\mathbf{x}})$,
- an $n \times n$ symmetric Hessian matrix $H(\bar{\mathbf{x}}) = \nabla^2 f(\bar{\mathbf{x}})$, and
- a function $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}$

such that for each $\mathbf{x} \in S$:

$$f(\mathbf{x}) = \underbrace{f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) + \frac{1}{2} (\mathbf{x} - \bar{\mathbf{x}})^T H(\bar{\mathbf{x}}) (\mathbf{x} - \bar{\mathbf{x}})}_{\text{Second-order Taylor series approximation of } f \text{ near } \bar{\mathbf{x}}} + \|\mathbf{x} - \bar{\mathbf{x}}\|^2 \alpha(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}).$$

Examples

1. $f(\mathbf{x}) = 4x_1 + 2x_2 - 7x_1^2 - 3x_2^2 + 5x_1x_2$ at $\bar{\mathbf{x}} = \mathbf{0}$.
2. $f(\mathbf{x}) = e^{2x_1+3x_2}$ at $\bar{\mathbf{x}} = (2, 1)$.

Roadmap

The Constrained Optimization Problem

The Lagrangian Function

KKT Conditions

KKT Necessary Conditions

KKT Sufficient Conditions

Constraint Qualifications

Review: Calculus Tools for KKT Proofs

Proof Space: Key Results

Applications and Examples

Proof Space: KKT Necessary Conditions

Theorem 15.7 (KKT Necessary Conditions)

Suppose $\bar{\mathbf{x}}$ is a local minimum of the constrained problem and a constraint qualification holds at $\bar{\mathbf{x}}$. Then there exist $\bar{\boldsymbol{\lambda}} \geq \mathbf{0}$ and $\bar{\boldsymbol{\nu}}$ such that:

1. *Stationarity:* $\nabla f(\bar{\mathbf{x}}) + \sum_{i=1}^m \bar{\lambda}_i \nabla g_i(\bar{\mathbf{x}}) + \sum_{j=1}^p \bar{\nu}_j \nabla h_j(\bar{\mathbf{x}}) = \mathbf{0}$
2. *Primal feasibility:* $g_i(\bar{\mathbf{x}}) \leq 0, h_j(\bar{\mathbf{x}}) = 0$
3. *Dual feasibility:* $\bar{\lambda}_i \geq 0$
4. *Complementary slackness:* $\bar{\lambda}_i g_i(\bar{\mathbf{x}}) = 0$

Proof Outline: (to be done on board)

1. Apply Fritz John Necessary Conditions (Theorem 15.6) to obtain multipliers $\lambda_0, \bar{\boldsymbol{\lambda}}, \bar{\boldsymbol{\nu}}$ with $(\lambda_0, \bar{\boldsymbol{\lambda}}, \bar{\boldsymbol{\nu}}) \neq \mathbf{0}$
2. Show complementary slackness: $\bar{\lambda}_i g_i(\bar{\mathbf{x}}) = 0$ and $\bar{\lambda}_i \geq 0$
3. Use constraint qualification (e.g., LICQ from Definition 15.7) to prove $\lambda_0 \neq 0$ (cannot have $\lambda_0 = 0$)
4. Normalize by dividing by λ_0 to obtain KKT form (Theorem 15.7)

See instructor proof notes for complete details.

Proof Space: KKT Sufficient Conditions (Convex Case)

Theorem 15.11 (KKT Sufficient Conditions for Convex Optimization)

Suppose f and $g_1, \dots, g_m$ are convex, $h_1, \dots, h_p$ are affine, and $(\bar{\mathbf{x}}, \bar{\boldsymbol{\lambda}}, \bar{\nu})$ satisfy the KKT conditions. Then $\bar{\mathbf{x}}$ is a global minimum of the constrained problem.

Proof Outline: (to be done on board)

1. Use convexity: $f(\mathbf{y}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{y} - \bar{\mathbf{x}})$ for any $\mathbf{y}$
2. Apply KKT stationarity to express $\nabla f(\bar{\mathbf{x}})$ in terms of multipliers
3. Use convexity of g_i and complementary slackness:
$$-\bar{\lambda}_i \nabla g_i(\bar{\mathbf{x}})^T(\mathbf{y} - \bar{\mathbf{x}}) \geq -\bar{\lambda}_i g_i(\mathbf{y}) \geq 0$$
4. Use affinity of h_j and feasibility: $-\bar{\nu}_j \nabla h_j(\bar{\mathbf{x}})^T(\mathbf{y} - \bar{\mathbf{x}}) = 0$
5. Combine: $f(\mathbf{y}) \geq f(\bar{\mathbf{x}}) + (\text{nonneg.}) + 0 \geq f(\bar{\mathbf{x}})$ for all feasible $\mathbf{y}$

See instructor proof notes for complete details.

Roadmap

The Constrained Optimization Problem

The Lagrangian Function

KKT Conditions

KKT Necessary Conditions

KKT Sufficient Conditions

Constraint Qualifications

Review: Calculus Tools for KKT Proofs

Proof Space: Key Results

Applications and Examples

Example: Least Squares with Nonnegativity

Problem

$$\min \|A\mathbf{x} - \mathbf{b}\|_2^2 \quad \text{s.t.} \quad \mathbf{x} \geq \mathbf{0}$$

Lagrangian: $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \|A\mathbf{x} - \mathbf{b}\|_2^2 - \boldsymbol{\lambda}^T \mathbf{x}$ (constraints: $-x_i \leq 0$)

KKT Conditions:

1. **Stationarity:** $2A^T(A\bar{\mathbf{x}} - \mathbf{b}) - \boldsymbol{\lambda}^* = \mathbf{0}$, i.e., $\boldsymbol{\lambda}^* = 2A^T(A\bar{\mathbf{x}} - \mathbf{b})$
2. **Primal feasibility:** $\bar{\mathbf{x}} \geq \mathbf{0}$
3. **Dual feasibility:** $\boldsymbol{\lambda}^* \geq \mathbf{0}$
4. **Complementary slackness:** $\lambda_i^* \bar{x}_i = 0$ for all i

Interpretation: For each i : either $\bar{x}_i = 0$ (at boundary) or $\lambda_i^* = 0$ (gradient component is zero).

Why KKT $\Rightarrow$ Optimality

The objective $f(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|_2^2$ is convex (positive semidefinite Hessian $2A^T A$), and the constraints are affine. So this is a convex problem and Slater's holds trivially (any $\mathbf{x} > \mathbf{0}$ is strictly feasible).

Therefore: **any KKT point is globally optimal**. The KKT conditions partition coordinates into:

- **Free variables** ($\bar{x}_i > 0$): $\lambda_i = 0$, so $[A^T(A\bar{\mathbf{x}} - \mathbf{b})]_i = 0$ (normal equations on these coords)
- **Boundary variables** ($\bar{x}_i = 0$): $\lambda_i = 2[A^T(A\bar{\mathbf{x}} - \mathbf{b})]_i \geq 0$ (residual gradient is nonneg.)

Example: Support Vector Machine (SVM)

Soft-Margin SVM

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

Convex QP with affine constraints $\Rightarrow$ KKT conditions are necessary and sufficient.

KKT stationarity (with multipliers $\alpha_i \geq 0$ for margin constraints, $\mu_i \geq 0$ for $\xi_i \geq 0$):

$$\mathbf{w} = \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i, \quad \sum_{i=1}^n \alpha_i y_i = 0, \quad C - \alpha_i - \mu_i = 0$$

Complementary slackness gives three cases for each data point i :

- $\alpha_i = 0$: point is correctly classified with margin > 1 (not a support vector)
- $0 < \alpha_i < C$: $\xi_i = 0$ and $y_i(\mathbf{w}^T \mathbf{x}_i + b) = 1$ (**support vector** on the margin)
- $\alpha_i = C$: $\xi_i \geq 0$, point is on or inside the margin (possibly misclassified)

Key insight: $\mathbf{w}$ is built from support vectors only ($\alpha_i > 0$). KKT determines the full solution.

Example: Projection onto a Simplex

Problem

Project a point $\mathbf{a} \in \mathbb{R}^n$ onto the standard simplex:

$$\min_{\mathbf{x}} \frac{1}{2} \|\mathbf{x} - \mathbf{a}\|^2 \quad \text{s.t.} \quad \sum_{i=1}^n x_i = 1, \quad \mathbf{x} \geq \mathbf{0}$$

KKT conditions (with ν for the equality, $\lambda_i \geq 0$ for $-x_i \leq 0$):

Stationarity: $x_i - a_i + \nu - \lambda_i = 0 \Rightarrow x_i = a_i - \nu + \lambda_i$

Comp. slackness: $\lambda_i x_i = 0$, so either $x_i > 0$ (then $\lambda_i = 0$, giving $x_i = a_i - \nu$) or $x_i = 0$.

Solution via KKT

$$\bar{x}_i = \max(a_i - \nu^*, 0) \quad \text{where } \nu^* \text{ satisfies } \sum_i \max(a_i - \nu^*, 0) = 1$$

Takeaway: KKT conditions yield a *closed-form algorithm*: sort, then find the threshold ν^* . This is widely used in optimization and machine learning (e.g., attention mechanisms, sparse inference).

Worked Example: Convex Problem with Case Analysis

Problem

Use KKT conditions to solve:

$$\min x_1^2 + x_2^2 \quad \text{s.t.} \quad -2x_1 - x_2 + 10 \leq 0, \quad x_2 \geq 0$$

Step 0: Convexity check. Objective is convex; constraints are affine.

Convex problem $\Rightarrow$ KKT sufficient.

Step 1: Lagrangian. Rewrite $x_2 \geq 0$ as $-x_2 \leq 0$:

$$\mathcal{L}(x_1, x_2, \lambda_1, \lambda_2) = x_1^2 + x_2^2 + \lambda_1(-2x_1 - x_2 + 10) + \lambda_2(-x_2)$$

Worked Example: Convex Problem (Solution)

Step 2: KKT conditions.

1. Stationarity: $\nabla_{\mathbf{x}} \mathcal{L} = \begin{pmatrix} 2x_1 - 2\lambda_1 \\ 2x_2 - \lambda_1 - \lambda_2 \end{pmatrix} = \mathbf{0}$
2. Comp. slackness: $\lambda_1(-2x_1 - x_2 + 10) = 0, \quad \lambda_2(-x_2) = 0$
3. Primal: $-2x_1 - x_2 + 10 \leq 0, \quad x_2 \geq 0$
4. Dual: $\lambda_1, \lambda_2 \geq 0$

Worked Example: Convex Problem (Solution)

Step 2: KKT conditions.

1. Stationarity: $\nabla_{\mathbf{x}} \mathcal{L} = \begin{pmatrix} 2x_1 - 2\lambda_1 \\ 2x_2 - \lambda_1 - \lambda_2 \end{pmatrix} = \mathbf{0}$
2. Comp. slackness: $\lambda_1(-2x_1 - x_2 + 10) = 0, \quad \lambda_2(-x_2) = 0$
3. Primal: $-2x_1 - x_2 + 10 \leq 0, \quad x_2 \geq 0$
4. Dual: $\lambda_1, \lambda_2 \geq 0$

Step 3: Case analysis on complementary slackness.

Case: $\lambda_1 = 0, \lambda_2 = 0$. Stationarity gives $x_1 = x_2 = 0$. But $-0 - 0 + 10 = 10 > 0$. **Infeasible.**

Case: $\lambda_1 \neq 0, \lambda_2 \neq 0$. Then $x_2 = 0$ and $-2x_1 + 10 = 0 \Rightarrow x_1 = 5$.

Stationarity: $\lambda_1 = 5, \lambda_2 = -5 < 0$. **Dual infeasible.**

Case: $\lambda_1 \neq 0, \lambda_2 = 0$. Then $-2x_1 - x_2 + 10 = 0$. Stationarity:

$x_1 = \lambda_1, x_2 = \lambda_1/2$. Substituting: $-2\lambda_1 - \lambda_1/2 + 10 = 0 \Rightarrow \lambda_1 = 4$. So $x_1 = 4, x_2 = 2, \lambda_1 = 4, \lambda_2 = 0$.

KKT point: $\bar{\mathbf{x}} = (4, 2), \boldsymbol{\lambda} = (4, 0)$. Since convex: **globally optimal**, $f^* = 20$.

Worked Example: Non-Convex Maximization

Problem

$$\max x_1^2 + 2x_1x_2 + 2x_2^2 - 3x_1 + x_2 \quad \text{s.t.} \quad x_1 + x_2 = 1, \quad x_1, x_2 \geq 0$$

(1) Is the problem convex? (2) Find all KKT points. (3) Find the optimal solution.

Convexity: Rewrite as $-\min -(x_1^2 + 2x_1x_2 + 2x_2^2 - 3x_1 + x_2)$.

The Hessian of the objective is $\begin{pmatrix} -2 & -2 \\ -2 & -4 \end{pmatrix}$ (for the minimization). **Not PSD $\Rightarrow$ not convex.**

Lagrangian (for $\min -f$): with ν for equality, $\lambda_1, \lambda_2 \geq 0$ for $-x_i \leq 0$:

$$\mathcal{L} = -(x_1^2 + 2x_1x_2 + 2x_2^2 - 3x_1 + x_2) + \nu(x_1 + x_2 - 1) - \lambda_1x_1 - \lambda_2x_2$$

Worked Example: Non-Convex Maximization (Solution)

KKT stationarity:

$$\nabla_{\mathbf{x}} \mathcal{L} = \begin{pmatrix} -2x_1 - 2x_2 + 3 + \nu - \lambda_1 \\ -2x_1 - 4x_2 - 1 + \nu - \lambda_2 \end{pmatrix} = \mathbf{0}$$

Case analysis: From $x_1 + x_2 = 1$, comp. slackness ($\lambda_1 x_1 = 0$, $\lambda_2 x_2 = 0$), and stationarity:

- **Try** $\lambda_1 = 0$, $\lambda_2 = 0$: Subtracting stationarity equations:
 $2x_2 - 4 + \lambda_1 - \lambda_2 = 0 \Rightarrow x_2 = 2$. But $x_1 + x_2 = 1$ gives
 $x_1 = -1 < 0$. **Infeasible.**
- **Try** $x_1 = 0$ ($\lambda_1 \geq 0$), $\lambda_2 = 0$: $x_2 = 1$. Stationarity: $\nu = \lambda_1 - 1$ and
 $\nu = 5$. So $\lambda_1 = 6 \geq 0$ ✓. KKT point: $(0, 1)$, $f = 3$.
- **Try** $\lambda_1 = 0$, $x_2 = 0$ ($\lambda_2 \geq 0$): $x_1 = 1$. Stationarity: $\nu = -1$ and
 $\lambda_2 = \nu + 1 = 0$. KKT point: $(1, 0)$, $f = -2$.

Worked Example: Non-Convex Maximization (Solution)

KKT stationarity:

$$\nabla_{\mathbf{x}} \mathcal{L} = \begin{pmatrix} -2x_1 - 2x_2 + 3 + \nu - \lambda_1 \\ -2x_1 - 4x_2 - 1 + \nu - \lambda_2 \end{pmatrix} = \mathbf{0}$$

Case analysis: From $x_1 + x_2 = 1$, comp. slackness ($\lambda_1 x_1 = 0$, $\lambda_2 x_2 = 0$), and stationarity:

- **Try** $\lambda_1 = 0$, $\lambda_2 = 0$: Subtracting stationarity equations:
 $2x_2 - 4 + \lambda_1 - \lambda_2 = 0 \Rightarrow x_2 = 2$. But $x_1 + x_2 = 1$ gives
 $x_1 = -1 < 0$. **Infeasible.**
- **Try** $x_1 = 0$ ($\lambda_1 \geq 0$), $\lambda_2 = 0$: $x_2 = 1$. Stationarity: $\nu = \lambda_1 - 1$ and
 $\nu = 5$. So $\lambda_1 = 6 \geq 0$ ✓. KKT point: $(0, 1)$, $f = 3$.
- **Try** $\lambda_1 = 0$, $x_2 = 0$ ($\lambda_2 \geq 0$): $x_1 = 1$. Stationarity: $\nu = -1$ and
 $\lambda_2 = \nu + 1 = 0$. KKT point: $(1, 0)$, $f = -2$.

Since not convex: KKT is necessary but **not sufficient**. Compare values:

$f(0, 1) = 3 > f(1, 0) = -2$. Optimal solution: $(x_1^*, x_2^*) = (0, 1)$, $f^* = 3$.

Lesson: For non-convex problems, you must find *all* KKT points and compare objective values.

Exercise: End-of-Lecture Assessment

Quick Assessment

1. True/False: For a convex optimization problem, every local minimum is a global minimum.
2. True/False: Complementary slackness says that $\lambda_i > 0$ implies the i -th constraint is active.
3. True/False: KKT conditions are always sufficient for optimality.
4. Write the complementary slackness condition for:
 $g_1(\mathbf{x}) = x_1 - 3 \leq 0$ with multiplier λ_1 .
5. What is Slater's condition for: $\min f(\mathbf{x})$ s.t. $g_1(\mathbf{x}) \leq 0, g_2(\mathbf{x}) \leq 0$?

Answers

1. **True.** This is a fundamental property of convex optimization.
2. **True.** $\lambda_i g_i(\bar{\mathbf{x}}) = 0$ with $\lambda_i > 0$ forces $g_i(\bar{\mathbf{x}}) = 0$ (active).
3. **False.** KKT conditions are sufficient only for *convex* problems. For nonconvex problems, they are only necessary (given a constraint qualification).
4. **Answer:** $\lambda_1(x_1 - 3) = 0$, meaning either $\lambda_1 = 0$ or $x_1 = 3$.
5. **Slater's condition:** There exists $\tilde{\mathbf{x}}$ such that $g_1(\tilde{\mathbf{x}}) < 0$ **and** $g_2(\tilde{\mathbf{x}}) < 0$ (both constraints strictly satisfied).

Summary and Next Steps

Key Takeaways

- The **Lagrangian** incorporates constraints into the objective via multipliers
- **KKT conditions**: stationarity, primal/dual feasibility, complementary slackness
- For **convex problems**: $\text{KKT} \Leftrightarrow \text{optimality}$ (with constraint qualification)
- **Slater's condition**: existence of a strictly feasible point

Next Time (Lecture 7): Duality Theory

- The Lagrangian dual problem
- Weak and strong duality
- Dual interpretation of KKT multipliers

Questions?